

Clinical Enzymology



Diagnostic significance of enzymes

1)Enzymes can act as diagnostic markers of underlying diseases .

2)Enzymes can also act as reagents for various biochemical estimations and detections

Enzymes as diagnostic markers

1)Functional plasma enzymes(Plasma derived enzymes)- Certain enzymes, proenzymes, and their substrates are present at all times in the circulation of normal individuals and perform a physiologic function in the blood.

Examples of these **functional plasma enzymes** include lipoprotein lipase, pseudo cholinesterase, and the proenzymes of blood coagulation and blood clot dissolution .The majority of these enzymes are synthesized in and secreted by the liver.

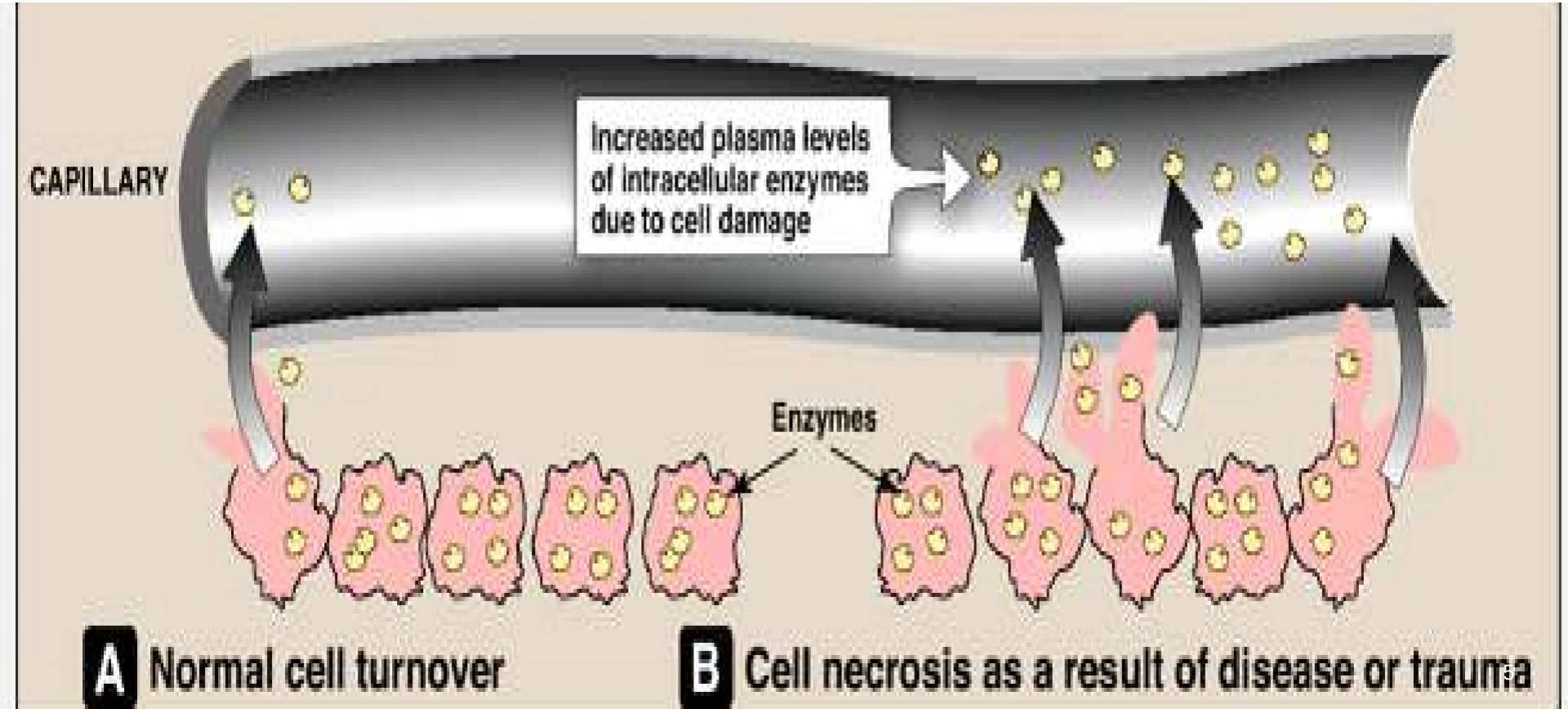
Enzymes as diagnostic markers

2) Nonfunctional plasma enzymes (Cell derived enzymes)-Plasma also contains numerous other enzymes that perform no known physiologic function in blood.

These apparently **nonfunctional plasma enzymes** arise from the routine normal destruction of erythrocytes, leukocytes, and other cells.

Enzymes as diagnostic markers

Tissue damage or necrosis resulting from injury or disease is generally accompanied by increases in the levels of several nonfunctional plasma enzymes.



Possible mechanisms responsible for increased serum levels

A) **Increased release**

i. Necrosis of cell

ii. Increased permeability of cell without gross cellular damage

iii. Increased production of enzyme within the cell resulting in increase in serum by overflow

iv. Increase in tissue source of enzyme as in malignancy

B) **Impaired disposition**

i. Increased levels in obstructive jaundice

ii. Increased levels in renal failure

Decreased serum levels

A. **Decreased formation which may be**
   i. **Genetic**
   ii. **Acquired**
A. **Enzyme inhibition**
B. **Lack of cofactors**

Units of serum enzyme activity

International unit – One IU is defined as the activity of the enzyme which transforms one micro mole of substrate in to products per minute per litre of sample under optimal conditions and at defined temperature . It is expressed as IU/L

Katal – catalytic unit – One Katal is defined as the number of mole of substrate transformed per second per litre of sample. It is abbreviated as kat or k.

60 U=mkat and 1nkat=0.06U

Clinical Significance of enzyme estimation

Single or serial assay of serum activity of a selected enzyme-

1) Helps in making the diagnosis/differential diagnosis/ early detection of a disease
2) Helps in ascertaining prognosis of a disease
3) Helps in ascertaining the response to drugs in a disease
4) Also help in ascertaining the time course of disease.

Enzymes as diagnostic markers in different diseases

Enzyme estimations are helpful in the diagnosis of –

1) Myocardial Infarction
2) Liver diseases
3) Muscle diseases
4) Bone diseases
5) Cancers
6) GI Tract diseases

Diagnosis of AMI



- The diagnosis of AMI is usually predicated on the WHO criteria of **chest pain, ECG changes, and increases in biochemical markers** of myocardial injury.
- Half of the patients with "typical" symptoms do not have AMI.
- The ECG is specific for AMI, but lacks sensitivity.
- In contrast, biochemical markers have excellent sensitivity for diagnosing AMI. By combining the most sensitive and the most specific tests, diagnostic accuracy can be enhanced.

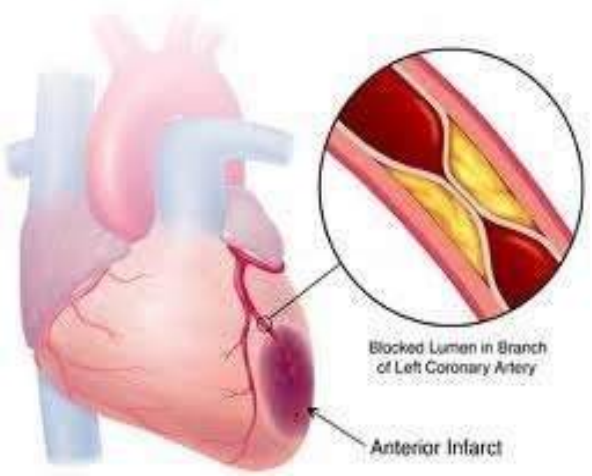
Serum enzymes in Acute Myocardial Infarction

Enzyme assays routinely carried out for the diagnosis of Acute Myocardial Infarction are-

1) **Creatine Phospho kinase,**

2) **Aspartate transaminase and**

3) **Lactate dehydrogenase**



Acceptable biochemical markers of ischemic heart disease are now considered to include **myoglobin, CK-MB, total CK, and cardiac troponins T and I.**

1) Creatine Kinase (CK, CPK)

- It is an enzyme found primarily in the heart and skeletal muscles, and to a lesser extent in the brain but not found at all in liver and kidney
- **Catalyzes the transfer of phosphate between creatine and ATP/ADP**
- **Provides rapid regeneration of ATP when ATP is low**
- **Creatine phosphate is regenerated when ATP is abundant**

Creatine Kinase (CK, CPK)

- **After myocardial infarction-** serum value is found to increase within 3-6 hours, reaches a peak level in 24- 30 hours and returns to normal level in 2-4 days (usually in 72 hours).
- **Normal Value**- serum activity varies from 10-50 IU/L at 30°C.
- CK is a **sensitive indicator** in the early stages of myocardial ischemia.
- No increase in activity is found in heart failure and coronary insufficiency.
- In acute MI, CPK usually rises faster than SGOT and returns to normal faster than the SGOT.

CK/CPK Isoenzymes

- There are three Isoenzymes.
- Measuring them is of value in the presence of elevated levels of CK or CPK to determine the source of the elevation.
- Each iso enzyme is a dimer composed of two protomers ‘M’ (for muscles) and ‘B’(for Brain).
- These isoenzymes can be separated by, Electrophoresis or by Ion exchange Chromatography.

CK/CPK Isoenzymes

Isoenzyme	Electrophoretic mobility	Tissue of origin	Mean percentage in blood
MM(CK3)	Least	Skeletal muscle Heart muscle	97-100%
MB(CK2)	Intermediate	Heart muscle	**0-3%**
BB(CK1)	Maximum	Brain	**0%**

CK/CPK Isoenzymes

- Normal levels of CK/CPK are almost entirely MM, from skeletal muscle.
- Elevated levels of CK/CPK resulting from acute myocardial infarction are about half MM and half MB.
- Myocardial muscle is the only tissue that contains more than five percent of the total CK activity as the CK2 (MB) isoenzyme.
- **Following an attack of acute myocardial infarction, this isoenzyme appears within 4 hours following onset of chest pain, reaches a peak of activity at approximately 24 hours and falls rapidly.**
- MB accounts for 4.5-20% of the total CK activity in the plasma of the patients with recent myocardial infarction and the total isoenzyme is elevated up to 20-folds above the normal.

2) Aspartate amino Transferase (AST)

- It is also called as Serum Glutamate Oxalo acetate Transaminase (SGOT).
- The level is significantly elevated in Acute MI.
- **Normal Value-** 0-41 IU/L at 37°C
- **In acute MI-** Serum activity rises sharply within the first 12 hours, with a peak level at 24 hours or over and returns to normal within 3-5 days.
- The rise depends on the extent of infarction. Re- infarction results in secondary rise of SGOT.

Aspartate amino Transferase (AST)

Prognostic significance- Levels> 350 IU/L are due to massive infarction (Fatal), > 150 IU/L are associated with high mortality and levels < 50 IU/L are associated with low mortality.

Other diseases- The rise in activity is also observed in muscle and hepatic diseases. These can be well differentiated from simultaneous estimations of other enzyme activities like SGPT etc, which do not show and rise in activity in Acute MI.

3) Lactate dehydrogenase (LDH)

- Lactate dehydrogenase catalyzes the reversible conversion of pyruvate and lactate.
- **Normal level-** 55-140 IU/L at 30°C. The levels in the upper range are generally seen in children.
- LDH level is 100 times more inside the RBCs than in plasma, and therefore minor amount of hemolysis results in false positive result.

Lactate dehydrogenase (LDH)

- **In Acute MI-**The serum activity rises within 12 to 24 hours, attains a peak at 48 hours (2 to 4 days) reaching about 1000 IU/L and then returns gradually to normal from 8 th to 14 th day.
- The magnitude of rise is proportional to the extent of myocardial infarction.
- Serum LDH elevation may persist for more than a week after CPK and SGOT levels have returned to normal levels.

Lactate dehydrogenase (LDH)

Other diseases-The increase in serum activity of LDH is also seen in hemolytic anemias, hepatocellular damage, muscular dystrophies, carcinoma, Leukemias, and any condition which causes necrosis of the body cells.

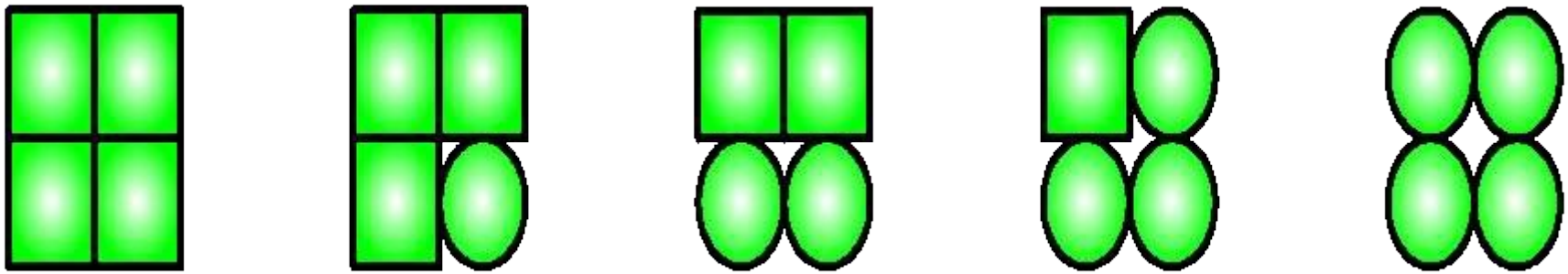
Since the total LDH is increased in many diseases, so the study of Iso enzymes of LDH is of more significance.

Iso enzymes of LDH

- LDH enzyme is tetramer with 4 subunits.
- The subunit may be either H(Heart) or M(Muscle) polypeptide chains.
- These two chains are the product of 2 different genes.
- Although both of them have the same molecular weight, there are minor amino acid variations.
- There can be 5 possible combinations;

H4, H3M1, H2M2, H1M3. M4, these are 5 different types of isoenzymes seen in all individuals.

Iso enzymes of LDH



LDH_1	LDH_2	LDH_3	LDH_4	LDH_5
H_4	H_3M_1	H_2M_2	H_1M_3	M_4

H subunit

M subunit

Iso enzymes of LDH

No. of Isoenzyme	Subunit make up of isoenzyme	Electrophoretic mobility at pH8.6	Activity at 60°for 30 minutes	Tissue origin	Percentage in human serum (Mean)
LDH-1	H4	Fastest	Not destroyed	Heart muscle	30%
LDH-2	H3M1	Faster	Not destroyed.	RBC	35%
LDH-3	H2M2	Fast	Partly destroyed	Brain	20%
LDH-4	H1M3	Slow	Destroyed	Liver	10%
LDH-5	M4	Slowest	Destroyed	Skeletal Muscles	5%

Normally LDH- 2(H3M1) level in blood is greater than LDH-1, but this pattern is reversed in myocardial infarction, this is called **'flipped pattern'**.

4) Cardiac Troponins

- They are not enzymes; however they are accepted as **markers of myocardial infarction.**
- The Troponin complex consists of 3 components; **Troponin C (Calcium binding), Troponin I(Actomyosin ATPase inhibitory element), and Troponin T(Tropomyosin binding element).**
- **Troponin I** is released in to the circulation within 4 hours of the onset of cardiac manifestations, peak is observed at 14-24 hours and remains elevated for 3-5 days post infarction.
- Serum level of **TnT increases within 6 hous of myocardial infarction, peaks at 72 hours and then remains elevated up to 7-10 days.** The TnT2 estimation is 100% sensitive index for myocardial infarction.

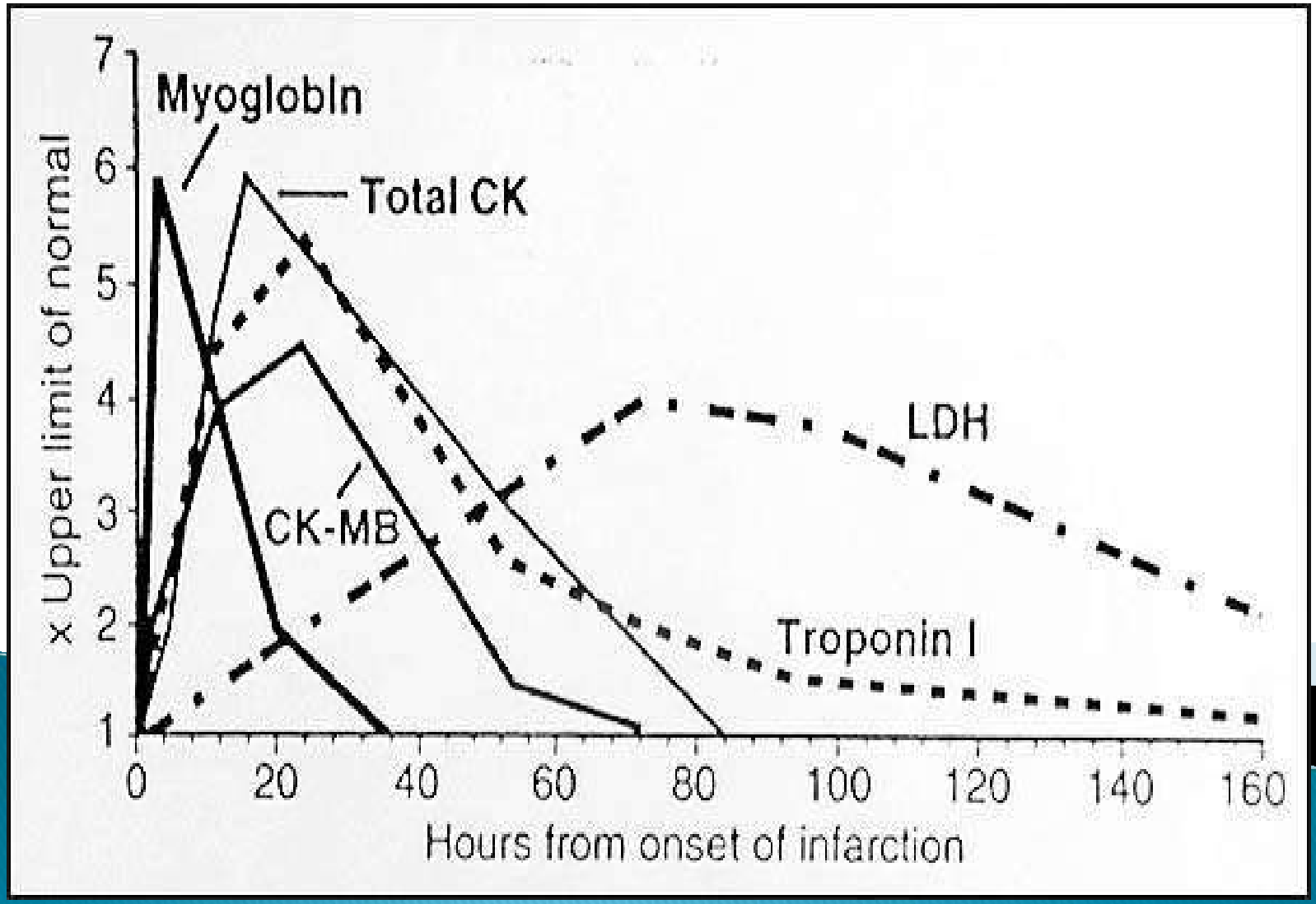
5) Myoglobin as cardiac marker

One of earliest markers is myoglobin, which is very sensitive but, in certain clinical settings, lacks specificity.

Its level rises within 4 hours of infarction

Falsely high levels may be observed in patients of Renal failure or patients having muscle injuries.

Enzymatic activity changes in Acute MI



Serum enzymes in liver diseases

Serum enzyme tests can be grouped into two categories:

(1) enzymes whose elevation in serum reflects damage to hepatocytes

(2) enzymes whose elevation in serum reflects cholestasis.

(1) Enzymes that Reflect Damage to Hepatocytes

The aminotransferases (transaminases) are sensitive indicators of liver cell injury and are most helpful in recognizing acute hepatocellular diseases such as hepatitis. These include-

1) Aspartate aminotransferase (AST) and
2) Alanine aminotransferase (ALT).

Amino Transferases

- **AST** is found in the liver, cardiac muscle, skeletal muscle, kidneys, brain, pancreas, lungs, leukocytes, and erythrocytes in decreasing order of concentration.
- **Normal level- 0-41 IU/L**
- **ALT** is found primarily in the liver.
- **Normal level-0-45 IU/L**
- The aminotransferases are normally present in the serum in low concentrations. These enzymes are released into the blood in greater amounts when there is damage to the liver cell membrane resulting in increased permeability.

Diagnostic significance of Aminotransferases

- **Levels of up to 300 U/L are nonspecific and may be found in any type of liver disorder.**
- Striking elevations—i.e., **aminotransferases > 1000 U/L— occur almost exclusively in disorders associated with extensive hepatocellular injury such as (1) viral hepatitis, (2) ischemic liver injury (prolonged hypotension or acute heart failure), or (3) toxin- or drug-induced liver injury.**
- In most acute hepatocellular disorders, the ALT is higher than or equal to the AST.
- **An AST:ALT ratio > 2:1 is suggestive while a ratio > 3:1 is highly suggestive of alcoholic liver disease.**
- The AST in alcoholic liver disease is rarely >300 U/L and the ALT is often normal.
- A low level of ALT in the serum is due to an alcohol-induced deficiency of Pyridoxal phosphate.
- In obstructive jaundice the aminotransferases are usually not altered but ALP levels are abnormal.

(2) Enzymes that reflect Cholestasis

The activities of three enzymes—

1)Alkaline phosphatase,

2) 5'-nucleotidase, and

3)y-Glutamyl transpeptidase (GGT)—are usually elevated in cholestasis.

Alkaline phosphatase and 5'-nucleotidase are found in or near the bile canalicular membrane of hepatocytes, while GGT is located in the endoplasmic reticulum and in bile duct epithelial cells.

Alkaline phosphatase in Liver diseases

The normal serum alkaline phosphatase consists of many distinct isoenzymes found in the
liver, bone, placenta, and, less commonly, small intestine.

Physiological Variations-

- Patients over age 60 can have a mildly elevated alkaline phosphatase.
- Individuals with blood types O and B can have an elevation of the serum alkaline phosphatase after eating a fatty meal due to the **influx of intestinal alkaline phosphatase** into the blood.
- It is also nonpathologically elevated in children and adolescents undergoing rapid bone growth, because of bone alkaline phosphatase.
- It is also high late in normal pregnancy due to the influx
of placental alkaline phosphatase.

Alkaline phosphatase in Liver diseases

Pathological variations-

- Elevation of liver-derived alkaline phosphatase is not totally specific for cholestasis, and a less than threefold elevation can be seen in almost any type of liver disease.
- Alkaline phosphatase elevations greater than four times normal occur primarily in patients with **cholestatic liver disorders, infiltrative liver diseases such as cancer and amyloidosis.**

Alkaline phosphatase in Liver diseases

- In liver diseases, the elevation is almost always due to increased amounts of the liver isoenzyme.

- In the absence of jaundice or elevated aminotransferases, an elevated alkaline phosphatase of liver origin often, but not always, suggests **early cholestasis and, less often, hepatic infiltration by tumor or granulomata.**

- **Intrahepatic cholestasis**- Values are increased in drug-induced hepatitis, primary biliary cirrhosis, rejection of transplanted liver, and, rarely, alcohol-induced steatonecrosis.

- **Extrahepatic cholestasis**- Very high values are found in obstructive jaundice due to cancer, common duct stone, sclerosing cholangitis, or bile duct stricture

- The level of serum alkaline phosphatase elevation is not helpful in distinguishing between intrahepatic and extrahepatic cholestasis.

Isozymes of alkaline phosphatase

1. **Hepatic Isoenzyme** – Travels fastest towards the anode and occupies the same position as Alpha 2 globulin. Its level rises in extra hepatic biliary obstruction.
2. **Bone Isoenzyme**-Increases die to osteoblastic activity and is normally elevated in children during periods of active growth .
3. **Placental Isoenzyme** - Rises during last 6 weeks of pregnancy.
4. **Intestinal Isoenzyme-** Rise occurs after a fatty meal. May increase during various GI disorders.

Isozymes of alkaline phosphatase

Regan isoenzyme –

- **Present in plasma of about 15% of patients with carcinoma of lung, liver or gut**
- **Also seen in chronic smokers**
- **Structurally resembles placental ALP**

Nagao Isoenzyme

- **Variant of Regan isoenzyme**
- **Detected in metastatic carcinoma of pleural surfaces and adenocarcinoma of pancreas and bile duct.**

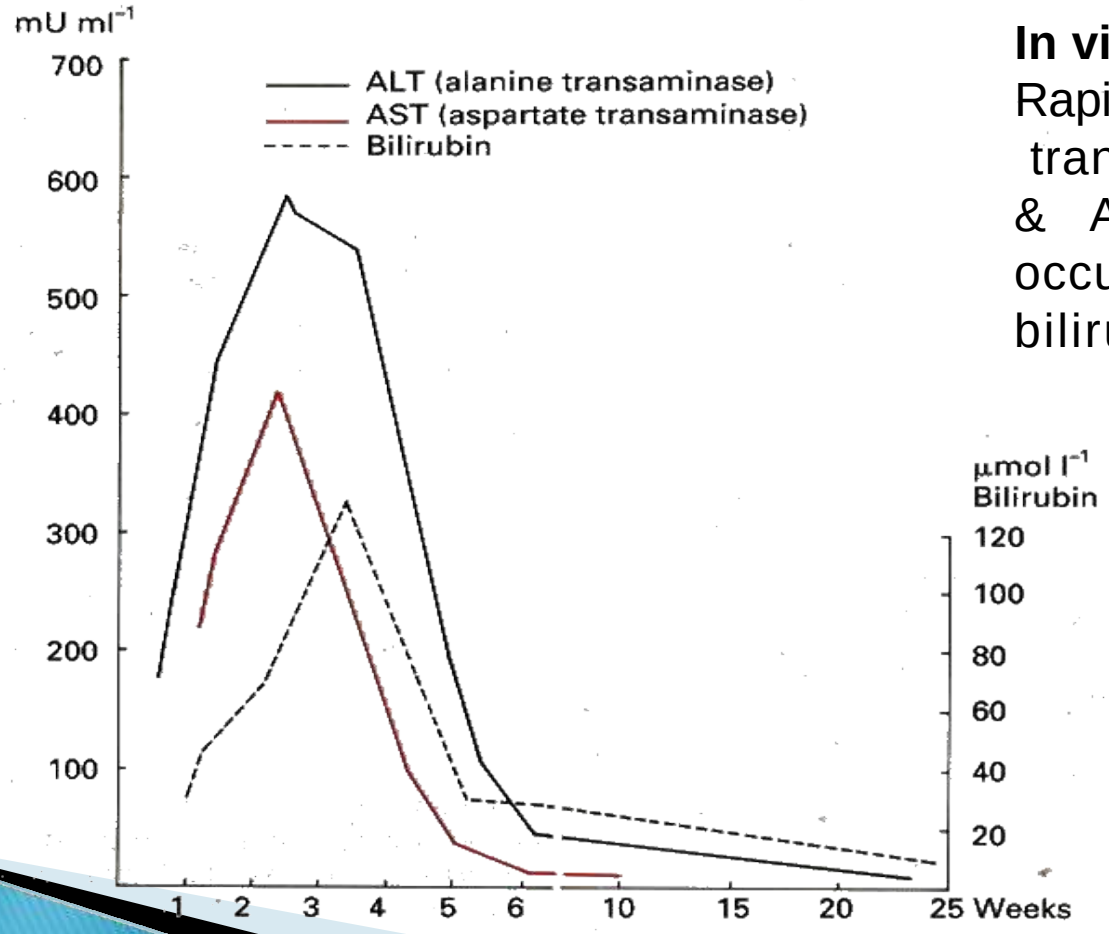
γ- Glutamyl transferase (γ GT)

- It is involved in **amino acid transport** across the membranes.
- Found mainly in **biliary ducts of the liver**, kidney and pancreas.
- Enzyme activity is induced by a **number of drugs** and in particular **alcohol.**
- γ GT increased in liver diseases especially in obstructive jaundice.
- γ GT levels are used as a marker of alcohol induced liver disease and in liver cirrhosis.

5’ Nucleotidase

- Moderately increased in hepatitis and highly elevated in biliary obstruction.
- Unlike ALP the level is unrelated to osteoblastic activity and is thus unaffected by bone disease.
- The enzyme hydrolyses 5’ nucleotides to 5’ nucleosides at an optimum p H of7.5

Serum enzymes in liver diseases



In viral hepatitis

Rapid rise in transaminases (AST & ALT) in serum occurs even before bilirubin rise is seen

3) Serum enzymes in Bone diseases

1) **Alkaline Phosphatase**-Rises in Rickets, osteomalacia, hyperparathyroidism and in Paget's disease. Also rises in primary and secondary malignancies of bones.
2) **Acid Phosphatase-**Highly increased in bony metastasis of carcinoma prostate

4) Serum Enzymes in muscle disease

1. **Aspartate Transferase-** Not commonly done
2. **Aldolase-**Moderate increase in Dermatomyositis, muscular dystrophies, highest values are seen in Deuchenne type of muscular dystrophies
3. **CPK-** Elevated in neurogenic muscular dystrophies, highest values are seen in Deuchenne type of muscular dystrophies

5) Serum enzymes in GI tract diseases

- **Amylase**-Serum activity > 1000 units is seen within 24 hours in acute Pancreatitis, values are diagnostic. A raised serum activity is also seen in perforated peptic ulcer and intestinal obstruction.
- **Lipase**-Levels as high as 2800 U/l are seen in acute pancreatitis. Also reported high in perforated duodenal and peptic ulcers and intestinal obstruction.

6) Enzymes as tumor markers

Enzyme	Disease
Serum acid phosphatase	Cancer prostate
Serum Alkaline phosphatase	Metastasis in liver, jaundice due to carcinoma head of pancreas, osteoblastic metastasis in bones
Serum LDH	Advanced malignancies and Leukemias
B- Glucuronidase	Cancer of urinary bladder
Leucine Amino Peptidase (LAP)	Liver cell carcinoma
Neuron specific Enolase	Malignancies of nervous tissue and brain

Enzymes as diagnostic reagents

Enzyme	Used for testing
Urease	Urea
Uricase	Uric acid
Glucose oxidase	Glucose
Cholesterol oxidase	Cholesterol
Lipase	Triglycerides
Alkaline phosphatase	ELISA
Horse radish Peroxidase	ELISA
Restriction endonuclease	Recombinant DNA technology
Reverse transcriptase	Polymerase chain reaction

Enzymes as therapeutic agents

Enzyme	Therapeutic Application
Streptokinase/Urokinase	Acute MI, Pulmonary embolism, DVT(Deep vein thrombosis)
Trypsin, lipase and amylase	Pancreatic insufficiency
Asparaginase/Glutaminase	Acute lymphoblastic leukemias
Hyaluronidase	Enhanced local anesthesia and for easy diffusion of fluids
Papain	Anti inflammatory
Chymotrypsin	Pain killer and Anti inflammatory
Alpha- 1 Antitrypsin	Deficiency and Emphysema
Serratopeptidase	Pain killer and Anti inflammatory

Summary–Enzymes as diagnostic markers

NAME OF THE ENZYME	Conditions in which level of activity in serum is elevated
Aspartate Amino transferase (AST) Serum glutamate-oxaloacetate transaminase (SGOT)	Myocardial infarction, Liver disease especially with liver cell damage
Alanine Amino transferase (ALT) Serum glutamate-pyruvate transaminase (SGPT)	Liver disease especially with liver cell damage
Alkaline Phosphatase (ALP)	Liver disease- biliary obstruction Osteoblastic bone disease-rickets
Acid Phosphatase (ACP)	Prostatic carcinoma
γ glutamyl Transferase (γ GT)	Liver disorder like liver cirrhosis and alcoholism
Creatine kinase (CK)	Myocardial infarction and skeletal muscle disease(muscular dystrophy
Lactate Dehydrogenase (LDH)	Myocardial infarction, other diseases